NINETEEN VECTORS AND A CERTIFICATE OF THEIR PHASE-RETRIEVAL INJECTIVITY IN $\mathbb{C}^6$

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Abstract. We exhibit nineteen explicit vectors in $\mathbb{C}^6$, with Gaussian-integer entries, that do phase retrieval: the intensities $|\langle a_j, x\rangle|^2$, $j = 1, \dots, 19$, determine every $x \in \mathbb{C}^6$ up to a unimodular scalar. This gives $m_{\mathbb{C}}(6) \le 19 = 4d - 5$, one below the generic count $4d - 4$; combined with the lower bound $m_{\mathbb{C}}(6) \ge 18$ of Wang and Xu, the minimal measurement number in dimension six is 18 or 19. The claim is verified by an exact rational computation, archived and rerunnable: after an elementary reduction, phase retrieval is equivalent to the emptiness of the real zero set of a system of five quadratic equations in five unknowns, and the archived eliminant of that system is a squarefree integer polynomial of degree fourteen with no real roots.

1. Statement

Fix measurement vectors $a_1, \dots, a_m \in \mathbb{C}^d$ and let

$$\Phi_A(x) = \big(|\langle a_1, x\rangle|^2, \dots, |\langle a_m, x\rangle|^2\big), \qquad \langle a, x\rangle = \textstyle\sum_k \bar{a}_k x_k.$$

The family *does phase retrieval* if $\Phi_A(x) = \Phi_A(y)$ forces $y = \omega x$ with $|\omega| = 1$; $m_{\mathbb{C}}(d)$ denotes the minimal m for which such a family exists in $\mathbb{C}^d$. Generic families of $4d - 4$ vectors do phase retrieval [2], and Vinzant [4] exhibited $11 = 4 \cdot 4 - 5$ vectors doing phase retrieval in $\mathbb{C}^4$. In dimension six, Wang and Xu proved $m_{\mathbb{C}}(6) \ge 18$ [5, Theorem 5.2], while the best upper bound known has been the generic $4d - 4 = 20$; whether $19 = 4d - 5$ vectors can do phase retrieval in $\mathbb{C}^6$ has been open. We answer affirmatively.

Theorem 1. *The nineteen row vectors of Table 1 do phase retrieval in $\mathbb{C}^6$. Consequently $18 \le m_{\mathbb{C}}(6) \le 19$.*

2. Verification

The verification is a finite, exact computation; this section records the elementary reduction that produces the computed system, and the archived facts about that system.

Reduction to polynomials. Every row of Table 1 is a scalar multiple of a vector $a(w) := (1, \bar{w}, \bar{w}^2, \dots, \bar{w}^5)$: rows 1–11 are $a(r)$ with $r = -5, \dots, 5$, and rows 12–19 are $25^5\, a(\zeta s)$ with

$$\zeta = \tfrac{-7+24i}{25}, \qquad s \in \{-5, -4, -3, -2, -1, 1, 3, 4\},$$

as one checks by expanding the powers. Scaling a measurement vector by a nonzero constant scales the corresponding intensity by a positive constant, so it suffices to verify phase retrieval for the unscaled family $\{a(r)\} \cup \{a(\zeta s)\}$. For $x \in \mathbb{C}^6$ let $f_x(z) = \sum_{k=0}^{5} x_{k+1} z^k$, so that $\langle a(w), x\rangle = f_x(w)$; the map $x \mapsto f_x$ is a linear bijection onto the polynomials of degree ≤ 5. For $f(z) = \sum c_k z^k$ write $f^{\#}(z) = \sum \bar{c}_k z^k$; then $f^{\#}(r) = \overline{f(r)}$ for real r, $|f(z)|^2 = f(z) f^{\#}(\bar{z})$ in general, and $(f^{\#})^{\#} = f$.

For real polynomials U, V put

$$\mathcal{W}(U,V)(t) = \tfrac{1}{2i}\Big(U(\zeta t)V(\bar{\zeta}t) - U(\bar{\zeta}t)V(\zeta t)\Big).$$

Its values at real t are real, so its coefficients are real; if $\deg U, \deg V \le k$, the coefficient of t^m collects the terms $u_i v_j - u_j v_i$ with $i < j$, $i + j = m$, so $\deg \mathcal{W}(U,V) \le 2k - 1$ and $t \mid \mathcal{W}(U,V)$; the coefficient of

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A6 = [
 [1, -5, 25, -125, 625, -3125],
 [1, -4, 16, -64, 256, -1024],
 [1, -3, 9, -27, 81, -243],
 [1, -2, 4, -8, 16, -32],
 [1, -1, 1, -1, 1, -1],
 [1, 0, 0, 0, 0, 0],
 [1, 1, 1, 1, 1, 1],
 [1, 2, 4, 8, 16, 32],
 [1, 3, 9, 27, 81, 243],
 [1, 4, 16, 64, 256, 1024],
 [1, 5, 25, 125, 625, 3125],
 [9765625, 13671875+46875000j, -205859375+131250000j,
  -918203125-804375000j, 2575515625-5533500000j, 30166521875+4615575000j],
 [9765625, 10937500+37500000j, -131750000+84000000j,
  -470120000-411840000j, 1054931200-2266521600j, 9884965888+1512431616j],
 [9765625, 8203125+28125000j, -74109375+47250000j,
  -198331875-173745000j, 333786825-717141600j, 2345748741+358907112j],
 [9765625, 5468750+18750000j, -32937500+21000000j,
  -58765000-51480000j, 65933200-141657600j, 308905184+47263488j],
 [9765625, 2734375+9375000j, -8234375+5250000j,
  -7345625-6435000j, 4120825-8853600j, 9653287+1476984j],
 [9765625, -2734375-9375000j, -8234375+5250000j,
  7345625+6435000j, 4120825-8853600j, -9653287-1476984j],
 [9765625, -8203125-28125000j, -74109375+47250000j,
  198331875+173745000j, 333786825-717141600j, -2345748741-358907112j],
 [9765625, -10937500-37500000j, -131750000+84000000j,
  470120000+411840000j, 1054931200-2266521600j, -9884965888-1512431616j],
]
```

TABLE 1. The nineteen measurement vectors $a_1, \dots, a_{19}$ for $\mathbb{C}^6$ (rows; $\mathtt{j} = \sqrt{-1}$).

t^1 is $\sin\theta\,(u_1v_0 - u_0v_1)$ with $\theta = \arg\zeta$, $\sin\theta = \frac{24}{25}$. A direct expansion with $u = U + iV$, $u^\# = U - iV$ gives, for real s,

$$|u(\zeta s)|^2 - |u^\#(\zeta s)|^2 = u(\zeta s)u^\#(\bar\zeta s) - u(\bar\zeta s)u^\#(\zeta s) = 4\,\mathcal{W}(U,V)(s). \tag{1}$$

Two elementary facts. *First*: if $f, g \neq 0$ have degree ≤ 5 and $|f| = |g|$ at the eleven points $r = -5, \dots, 5$, then $ff^\# - gg^\#$, of degree ≤ 10, vanishes at eleven points, so $ff^\# = gg^\#$; writing $h = \gcd(f,g)$, $f = hu$, $g = hv$ with $\gcd(u,v) = 1$ and cancelling $hh^\#$ gives $uu^\# = vv^\#$, whence v divides $u^\#$ and u divides $v^\#$, so by degrees $v = \lambda u^\#$ with $|\lambda| = 1$: every such pair has the form

$$f = hu, \qquad g = \omega\, h\, u^\#, \qquad |\omega| = 1, \qquad \deg h + \deg u \leq 5. \tag{2}$$

Second: if real U, V satisfy $\mathcal{W}(U,V) \equiv 0$, they are linearly dependent. Indeed ζ^2 is not a root of unity — otherwise ζ would be one, making $\zeta + \zeta^{-1} = 2\operatorname{Re}\zeta = -\frac{14}{25}$ a rational algebraic integer, hence an integer — and for coprime U, V the identity $U(\zeta t)V(\bar\zeta t) = U(\bar\zeta t)V(\zeta t)$ forces $U(\zeta t) \mid U(\bar\zeta t)$ (a common root τ of $U(\zeta t), V(\zeta t)$ would make $\zeta\tau$ a common root of U, V), so $U(\bar\zeta t) = \lambda U(\zeta t)$ and the root multiset of U is invariant under multiplication by ζ^2, which has infinite order: $U = u\,t^p$, symmetrically $V = v\,t^q$, $\min(p,q) = 0$ by coprimality, and substituting back forces $\zeta^{2(p-q)} = 1$, so $p = q = 0$ and both are constants.

Proposition 2. *The unscaled family fails phase retrieval if and only if there exist real linearly independent U, V of degree ≤ 5 and a real $\gamma \neq 0$ with*

$$\mathcal{W}(U,V)(t) \;=\; \gamma\, t\,(t+5)(t+4)(t+3)(t+2)(t+1)(t-1)(t-3)(t-4). \tag{3}$$

Moreover every such pair satisfies $u_1v_0 - u_0v_1 \neq 0$.

Proof. ($\Leftarrow$) Set $u = U + iV$ and let x, y be the coefficient vectors of $u, u^\#$. Then $|u(r)| = |u^\#(r)|$ for all real r, and at the eight rotated points (1) and (3) give $|u(\zeta s)| = |u^\#(\zeta s)|$; so all nineteen measurements agree. If $u^\# = \omega u$ with $|\omega| = 1$, then $(1-\omega)U = i(1+\omega)V$ would make U, V dependent; hence y is not a unimodular multiple of x, and phase retrieval fails.

($\Rightarrow$) Let $f = f_x$, $g = f_y$ witness a failure. If $f = 0$, then g vanishes at eleven > 5 real points, so $g = 0 = f$, a contradiction; so $f, g \neq 0$ and (2) applies, with $u \nparallel u^\#$ (else g is a unimodular multiple of f), i.e. U, V independent for $u = U + iV$. Let $k = \deg u$, so $\deg h \leq 5 - k$. At each rotated point with $h(\zeta s) \neq 0$, dividing $|f(\zeta s)| = |g(\zeta s)|$ by $|h(\zeta s)|$ and using (1) gives $\mathcal{W}(U,V)(s) = 0$; since h has at most

$5-k$ roots, at least $8-(5-k)=k+3$ of the eight values s are roots of $\mathcal{W} := \mathcal{W}(U,V)$. Write $\mathcal{W} = t\widetilde{\mathcal{W}}$, $\deg \widetilde{\mathcal{W}} \le 2k-2$. If $k \le 4$, then $k+3 > 2k-2$: $\widetilde{\mathcal{W}}$ has more distinct nonzero roots than its degree, so $\mathcal{W} \equiv 0$ and the second fact above makes U, V dependent — a contradiction. Hence $k = 5$, h is constant, all eight equations survive, and $\widetilde{\mathcal{W}}$, of degree ≤ 8, has the eight listed roots: $\widetilde{\mathcal{W}} = \gamma \prod(t-s)$ with $\gamma \ne 0$ (else $\mathcal{W} \equiv 0$ again), which is (3). Comparing coefficients of t^1 in (3) gives $\frac{24}{25}(u_1v_0 - u_0v_1) = \gamma \prod_s(-s) \ne 0$. □

The computation. Equation (3) is linear in the Plücker coordinates $p_{ij} = u_iv_j - u_jv_i$ of the plane spanned by U, V. In the chart $u_1v_0 - u_0v_1$ normalized by the last clause of Proposition 2, the coefficient equations of (3) reduce to five quadratic equations in five unknowns with rational coefficients. The archived computation [3] certifies, in exact rational arithmetic with no floating point:

- the system is zero-dimensional, and its eliminant with respect to one coordinate is a squarefree integer polynomial of degree fourteen (matching the Bézout count for a proper linear section of the Plücker-embedded $\mathrm{Gr}(2,6)$, whose degree is $C_4 = 14$);
- the Sturm sequence of the eliminant has seven sign variations at $-\infty$ and seven at $+\infty$: the eliminant has *no real roots*. A real solution of the system would make its eliminated coordinate a real root; hence no real solution exists, no pair (U, V, γ) as in Proposition 2 exists, and Theorem 1 follows.

Protocol: the Gröbner-basis computation was performed with `msolve` [1] over $\mathbb{Q}$; the real-root count is independently re-derived from the archived eliminant by a Sturm-sequence script in integer arithmetic; the polynomial system was generated by two independently written emitters and compared as canonically normalized sets before solving; as a sanity check, the 19×36 real matrix of the lifted measurements $\bar{a}_j a_j^\top$ has rank 19 over $\mathbb{Q}$; and the same pipeline, run on configurations engineered to fail, reports the expected nonzero real-solution counts. The `msolve` input, the certificate data (eliminant, Sturm sequence, sign counts, SHA-256 hashes), and a standalone exact verifier are archived at [3] (directory `certificate/`); the verification reruns in minutes on a laptop.

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